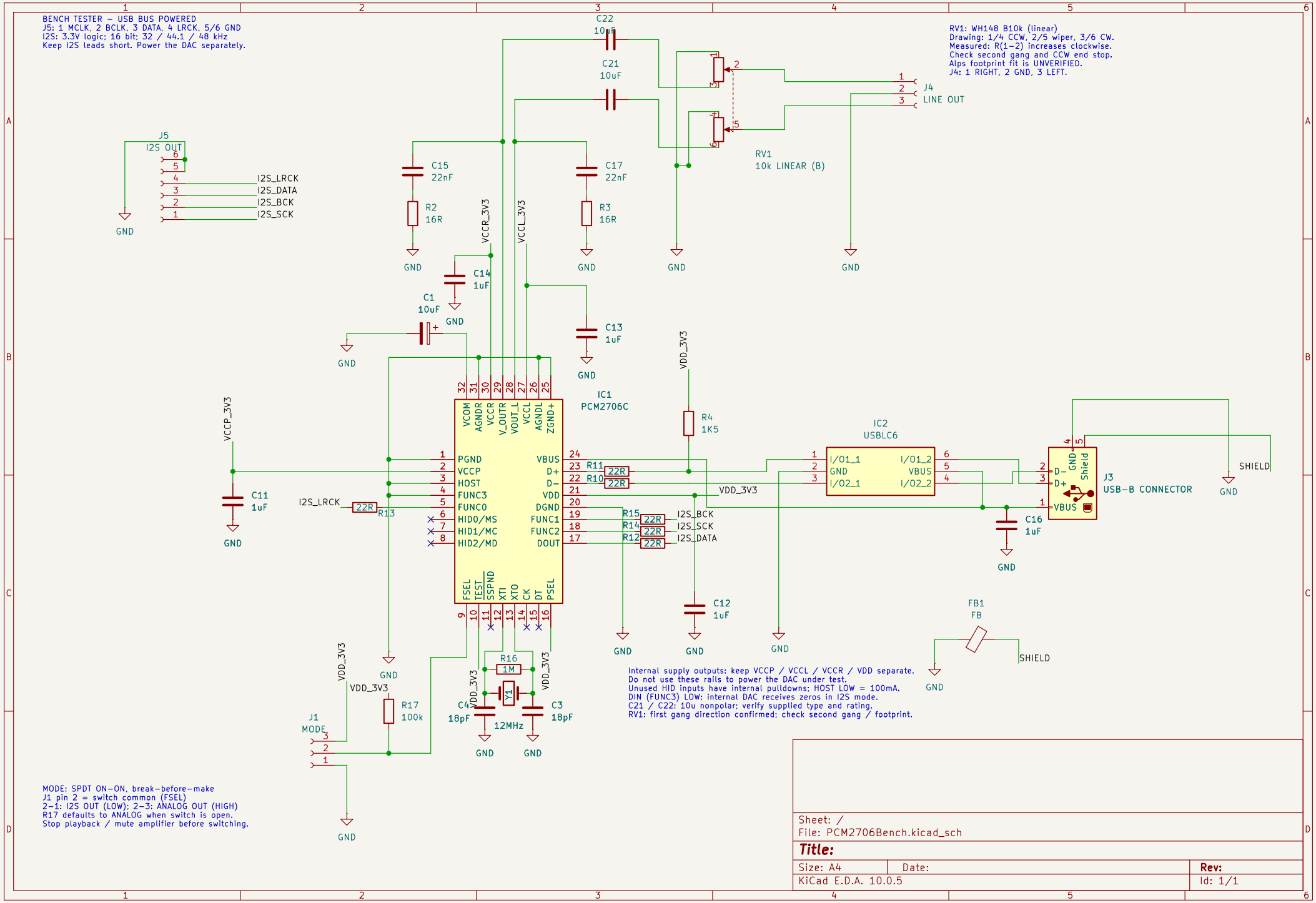

BENCH TESTER — USB BUS POWERED
J5: 1 MCLK, 2 BCLK, 3 DATA, 4 LRCK, 5/6 GND
I2S: 3.3V logic; 16 bit; 32 / 44.1 / 48 kHz
Keep I2S leads short. Power the DAC separately.
RV1: WH148 B10k (linear)
Drawing: 1/4 CCW, 2/5 wiper, 3/6 CW.
Measured: R(1–2) increases clockwise.
Check second gang and CCW end stop.
Alps footprint fit is UNVERIFIED.
J4: 1 RIGHT, 2 GND, 3 LEFT.
J4
LINE OUT
RV1
10k LINEAR (B)
J5
I2S OUT
I2S_LRCK
I2S_DATA
I2S_BCK
I2S_SCK
C22
10uF
C21
10uF
C15
22nF
C17
22nF
R2
16R
R3
16R
C14
1uF
C1
10uF
C13
1uF
VCCR_3V3
VCCL_3V3
VDD_3V3
IC1
PCM2706C
VCOM
AGNDR
VCCR
V_OUTR
VOUT_LL
VCCL
AGNDL
ZGND+
PGND
VCCP
HOST
FUNC3
FUNC0
HID0/MS
HID1/MC
HID2/MD
VBUS
D+
D–
VDD
DGND
FUNC1
FUNC2
DOUT
FSEL
TEST
SSPND
XTI
XTO
CK
DT
PSEL
R4
1K5
IC2
USBLC6
I/O1_1
GND
I/O2_1
I/O1_2
VBUS
I/O2_2
J3
USB–B CONNECTOR
D–
D+
VBUS
GND
Shield
SHIELD
C16
1uF
C11
1uF
VCCP_3V3
I2S_LRCK
R13
22R
R11
22R
R10
22R
R15
22R
R14
22R
R12
22R
I2S_BCK
I2S_SCK
I2S_DATA
C12
1uF
FB1
FB
R16
1M
Y1
12MHz
C4
18pF
C3
18pF
R17
100k
J1
MODE
Internal supply outputs: keep VCCP / VCCL / VCCR / VDD separate.
Do not use these rails to power the DAC under test.
Unused HID inputs have internal pulldowns; HOST LOW = 100mA.
DIN (FUNC3) LOW: internal DAC receives zeros in I2S mode.
C21 / C22: 10u nonpolar; verify supplied type and rating.
RV1: first gang direction confirmed; check second gang / footprint.
MODE: SPDT ON–ON, break–before–make
J1 pin 2 = switch common (FSEL)
2–1: I2S OUT (LOW); 2–3: ANALOG OUT (HIGH)
R17 defaults to ANALOG when switch is open.
Stop playback / mute amplifier before switching.
GND
Sheet: /
File: PCM2706Bench.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 1/1